

BTEC Level 3 Applied Science • Unit 1 Biology • Answer sheet

Microscopy — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

A–E

1. $M = \text{image size} / \text{actual size}$. Resolution: smallest distance at which two points can still be seen as separate.
2. $2500\ \mu\text{m}$; $2.5 \times 10^6\ \text{nm}$.
3. Eyepiece magnification $\times$ objective magnification.
4. $18\ \text{mm} = 18\ 000\ \mu\text{m}$; actual = $18\ 000/600 = 30\ \mu\text{m}$.
5. $3.0\ \text{cm} = 30\ 000\ \mu\text{m}$; $M = 30\ 000/7.5 = \times 4000$.
6. $40\ \text{mm} = 40\ 000\ \mu\text{m}$; $M = 40\ 000/10 = \times 4000$.
7. $35\ \text{mm} = 35\ 000\ \mu\text{m}$; actual = $70\ \mu\text{m}$. [2]
8. Light about $0.2\ \mu\text{m}$; TEM much smaller (about $0.1\ \text{nm}$) because electrons have a shorter wavelength. [4]
9. Living specimens / colour / cheaper / easier preparation. [2]
10. The granules are closer together than the resolution. Extra magnification enlarges the blur (empty magnification).